

# Adib Rouf

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## EDUCATION

### University of Connecticut, College of Engineering

Storrs, CT

*Bachelor of Science in Computer Science, Concentration in Artificial Intelligence*

Aug. 2024 – May 2027

- Relevant Coursework: Computer Architecture, Algorithms & Complexity, Data Structures, Object-Oriented Design, Discrete Systems, Systems Programming, Artificial Intelligence, Machine Learning

## EXPERIENCE

### Data Engineering Intern

May 2026 – August 2026

*The Travelers Companies*

Hartford, CT

- Modernized a legacy Active Directory workflow in Python and SQL, processing 200,000+ records across 8,500+ groups while achieving a **500% increase in processing speed** through parallelization
- Developed an AutoSys audit tool that identifies unmaintained jobs by tracing executing accounts, ownership, employment status, and line of business, improving job accountability and operational cleanup
- Developing a server health dashboard for proactive infrastructure monitoring, alerting employees to emerging issues before they become customer-facing incidents

### L'SPACE NPWEE Participant

Jan. 2026 – Apr. 2026

*NASA - National Aeronautics and Space Administration*

Remote

- Evaluated performance criteria for a proposed spaceflight materials testing protocol against mission-duration constraints, translating findings into **testable engineering requirements**
- Authored technical assumptions and requirement specifications incorporated into NASA's final proposal submission
- Collaborated with a multidisciplinary team to align testing methods, mission constraints, and feasibility considerations into a cohesive validation strategy

### Undergraduate Researcher

Mar. 2025 – Dec. 2025

*University of Connecticut — XAI4AV*

Storrs, CT

- Engineered a **full-stack ML pipeline** for uploading video and saliency data, triggering HPC-based LLM inference, and returning model decision heatmaps
- Designed an asynchronous status pipeline connecting the frontend, Supabase, and HPC compute backend, enabling reliable end-to-end job tracking across pending, processing, and complete states
- Delivered the upload, queueing, and results interface end-to-end, enabling researchers to compare AI-generated explanations with original video side by side

## PROJECTS

### ML Serving Platform | *Python, FastAPI, Async Job Queues*

June 2026 – Present

- Built an ML inference platform with FastAPI and asynchronous job queues, enabling non-blocking request processing through polling-based status tracking
- Implemented dynamic request batching in the worker pipeline, increasing inference throughput by **88% from 56 to 106 requests per second** under concurrent load
- Separated API request handling from model execution through a worker-based architecture, improving responsiveness and enabling the inference layer to scale independently

### Data Aggregation and Insights Platform | *Python, pandas, Snowflake*

Dec. 2024 – Mar. 2025

- Built a Python pipeline that transformed **28,900+ XML documents** into structured pandas DataFrames for downstream analysis
- Automated ingestion of cleaned data into Snowflake, creating a repeatable workflow for storage, querying, and analysis
- Improved Snowflake query performance by 30% by restructuring data-processing logic and optimizing record preparation

## TECHNICAL SKILLS

**Languages:** Python, SQL, C, JavaScript, HTML/CSS

**Frameworks & Tools:** FastAPI, Flask, React, Snowflake, Supabase, GitHub, Azure